

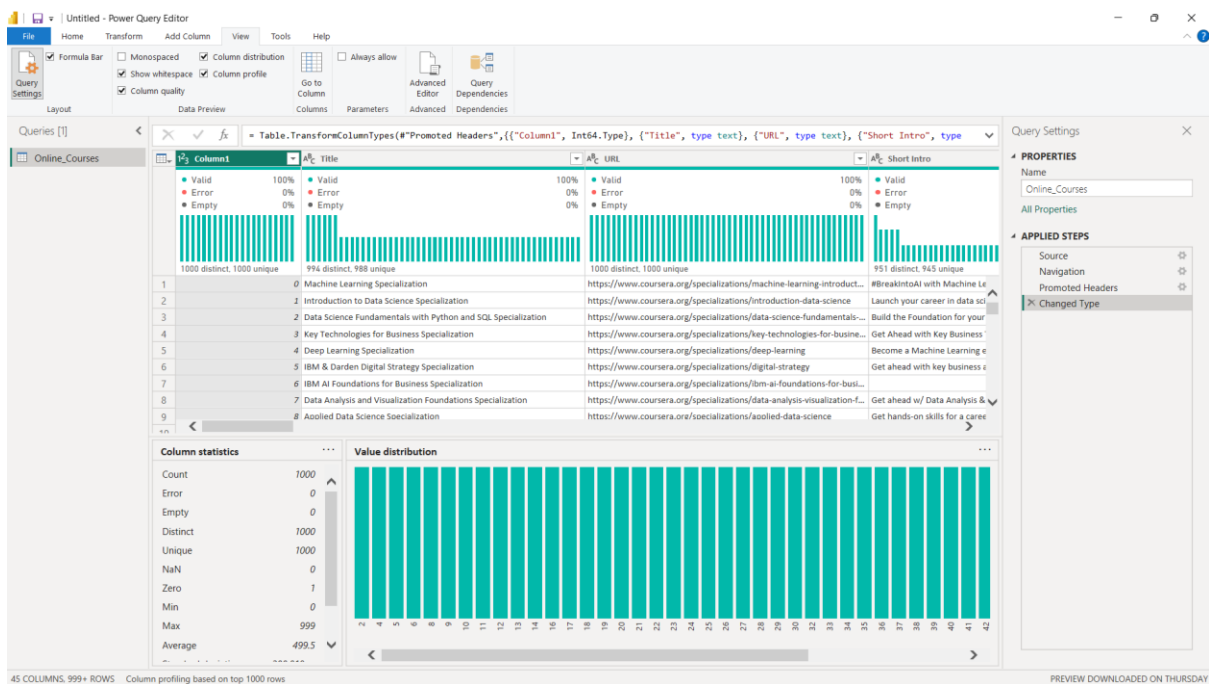
Part A: Data Understanding & Cleaning

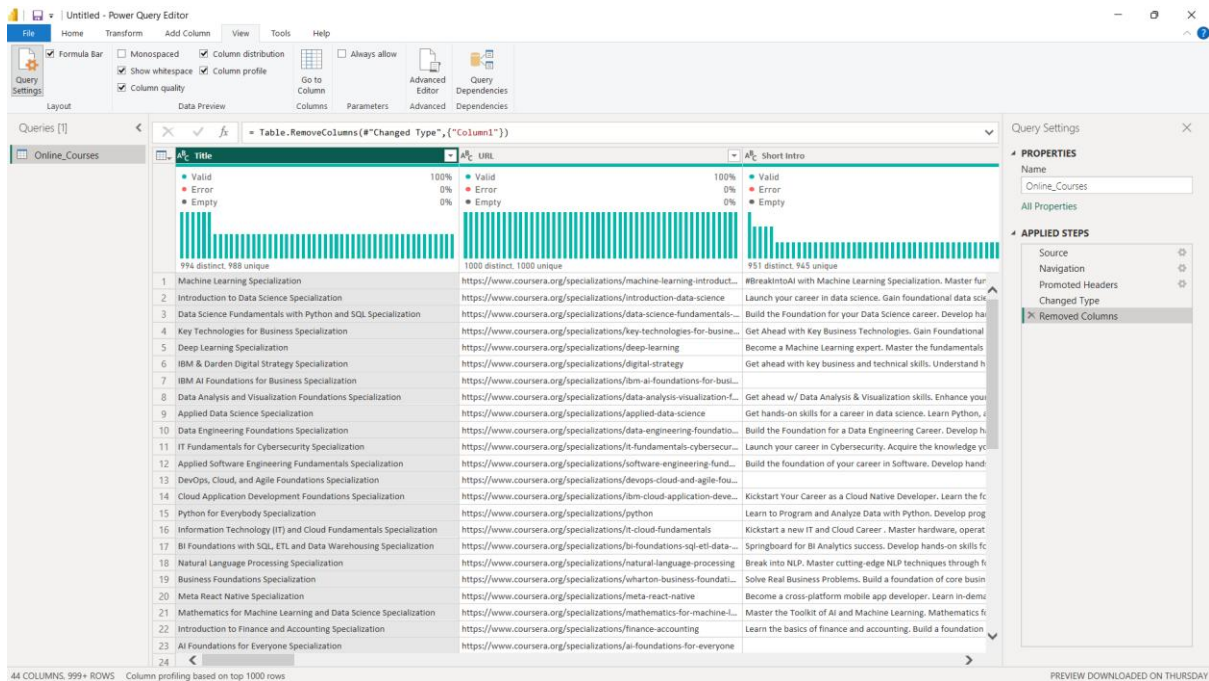
Question A1: Data Profiling

- An initial data profiling assessment was conducted on the raw scraped EdTech dataset using Power Query.
- The dataset consists of a total of 8,092 rows and 45 columns.
- While core fields such as Course Title, Category, and Language are completely populated, several scraped metadata columns contain high percentages of missing values, requiring systematic data cleaning before analysis.
- Below is the missing value analysis for the key fields required for the dashboard:
 - Category, Sub-category, Course Type, and Language were 0% missing (Valid).
 - Views, Rating, Instructor, Duration, Skills, and Subtitles required cleaning.

Question A2: Remove Irrelevant Columns

- The dataset was reviewed for irrelevant or non-value-adding fields.
- The auto-generated index field (Column1) was identified as providing no business value and was successfully removed from the dataset to streamline the model.





Question A3: Standardize Data Types

- **Set to Text (ABC):** Title / course title, Category, Sub category, Instructors, Language, Skills, Course type.
- **Set to Whole Number (123):** no of ratings (or No of reviews).
- **Set to Decimal Number (1.2):** price.

Question A4: Clean and Standardize "Views"

- A custom column named Views_Clean was created using conditional logic in Power Query to remove "k", "M", and commas, successfully converting the text-based views into a standardized numeric whole number format for aggregation.

Question A5: Clean Rating Column

- The original rating data was formatted as text strings containing the word "stars" (e.g., "4.5stars").
- To convert this into a usable numeric format, the 'Extract Text Before Delimiter' transformation was applied in Power Query using the delimiter "s".
- The column was successfully renamed to Rating_Clean and formatted as a Decimal Number data type to enable average rating calculations across categories.

Question A6: Create Subtitle Availability Column

- To analyze the impact of accessibility options on course viewership, a binary transformation was performed on the subtitle data.
- A custom column named Subtitle_Available was created using conditional M logic.

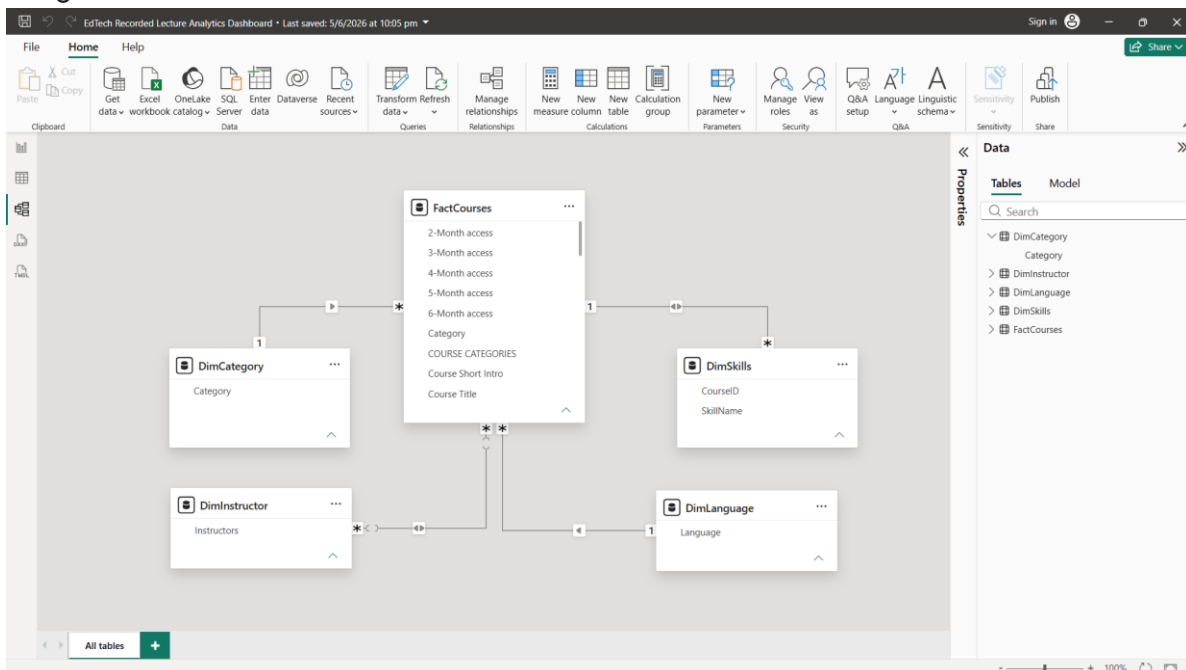
- Following the project's business rules, any rows where the original subtitle languages field was null, blank, or explicitly indicated "No" were flagged as "No"; all other populated rows were designated as "Yes".

Question A7: Convert Course Duration into Hours

- Course durations were provided in inconsistent text formats (e.g., months, hours, flexible schedules).
- A new numeric column, Duration_Hours, was created using conditional logic to standardize this data.
- Following the provided business rules, durations in months were extracted and multiplied by 60, "Flexible" schedules were assigned a baseline of 200 hours, and existing hourly durations were extracted and retained.
- The final column was formatted as a Decimal Number to support continuous variable analysis.

Part B: Data Modeling Question B1: Define the Star Schema Approach

- A Star Schema data model was implemented to ensure optimal dashboard performance and accurate filtering.
- The main dataset was established as the central fact table (FactCourses).
- Dimension tables (DimCategory, DimLanguage, DimInstructor) were created by referencing the fact table and removing duplicates to create unique lookup tables.
- A one-to-many relationship was established between the dimension tables and the fact table utilizing their respective text keys, while the DimSkills table was connected via a generated CourseID.



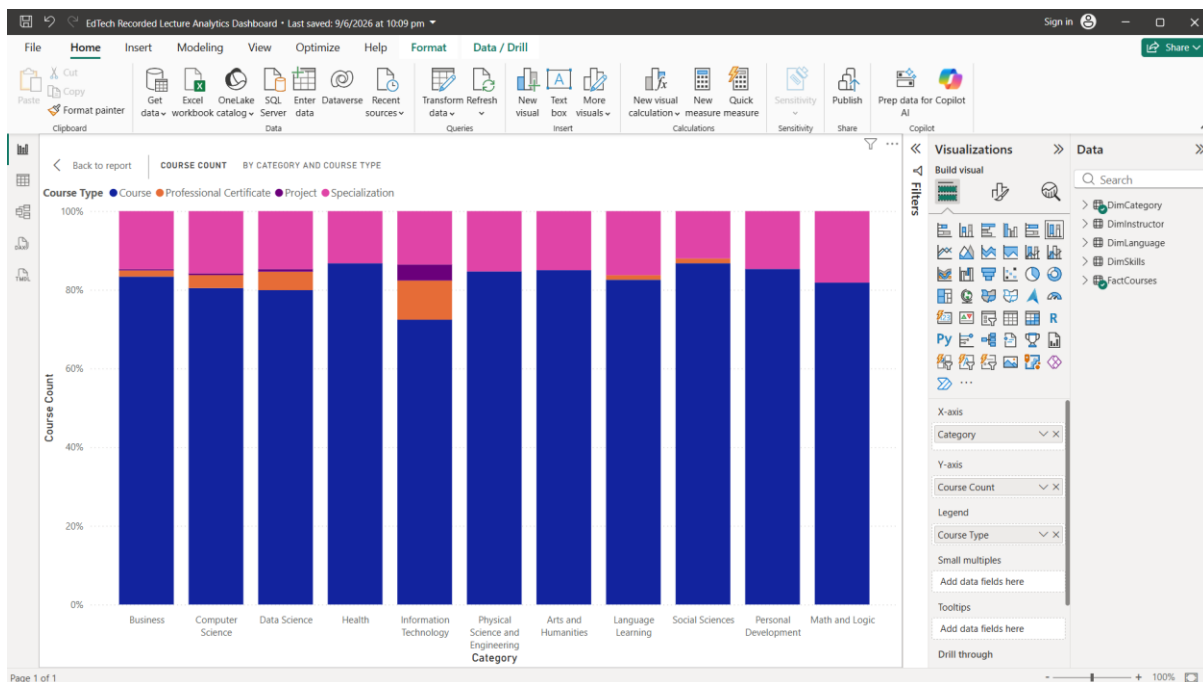
Question B2: Normalize Skills into a Skill Table

- To properly analyze skill variety and demand, the comma-separated Skills column was normalized into a row-level structure.
- A separate DimSkills table was created referencing the fact table.
- The Skills column was split by delimiter (comma) into separate rows using Power Query's advanced split options.
- Leading and trailing whitespaces were removed using the Trim transformation, resulting in a clean bridge table containing only CourseID and SkillName.

Part C: Dashboard Insights

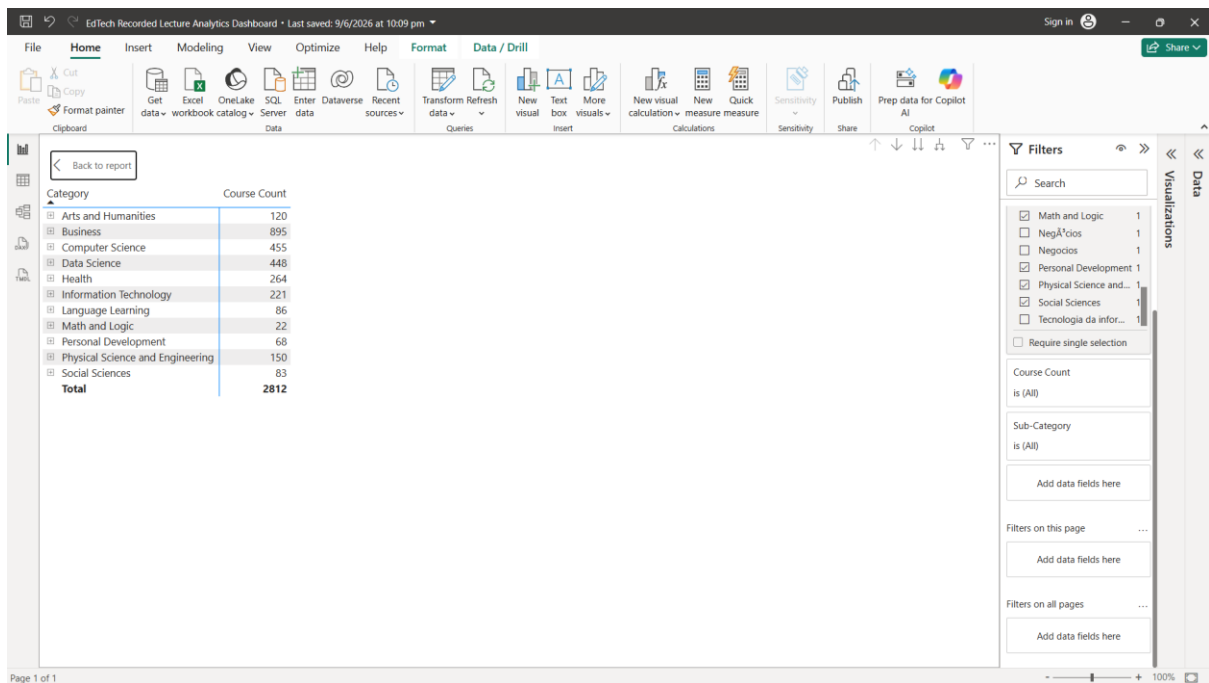
Question C1: Course Type Distribution by Category

- To analyze the composition of course offerings, a DAX measure was created (Course Count = COUNTROWS(FactCourses)) to accurately tally the number of courses.
- A 100% Stacked Column Chart was utilized to visualize the proportional distribution of Course Types across each main Category.



Question C2: Course Count by Category and Sub-category

- A matrix visual was created to display the total number of courses hierarchically.
- Using the previously created Course Count DAX measure, the matrix allows users to drill down from high-level Categories into specific Sub-categories to identify areas with the highest volume of course content.

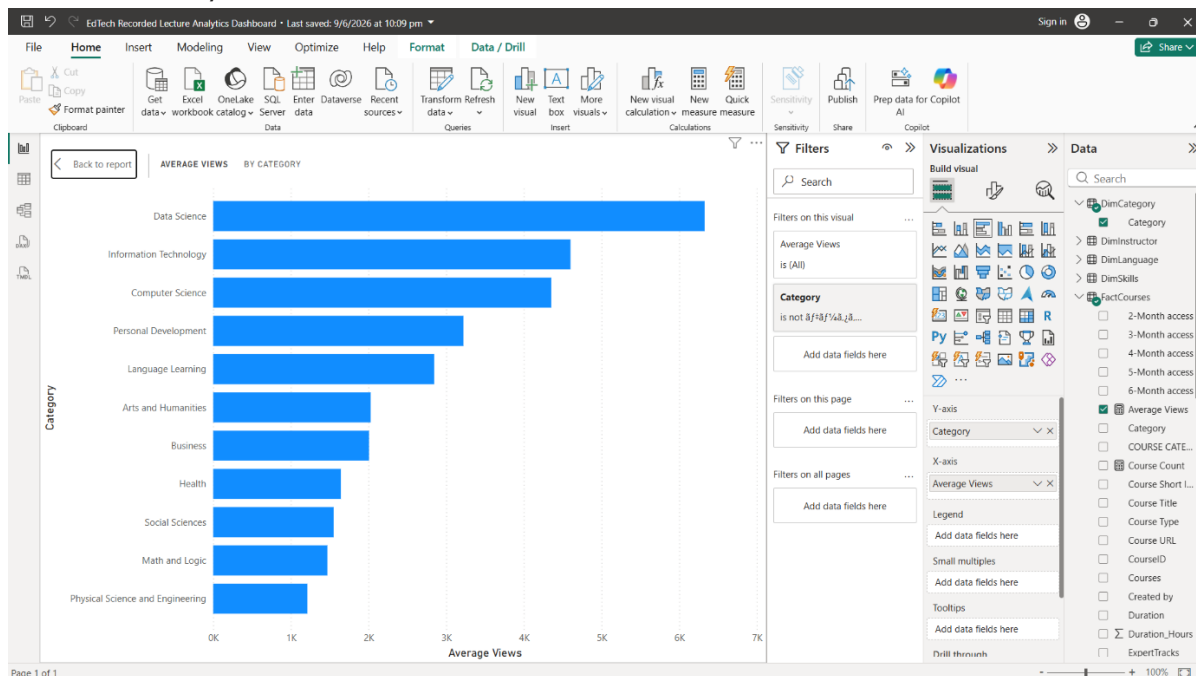


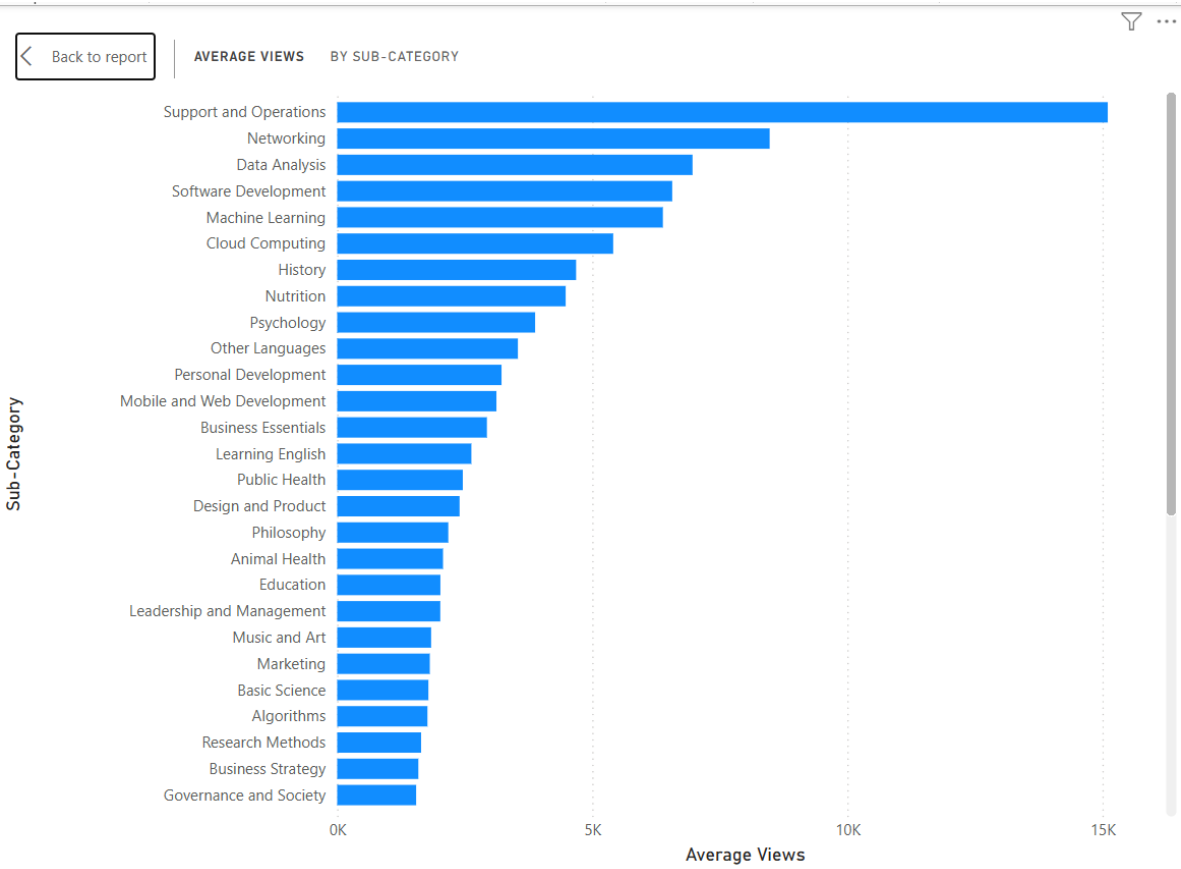
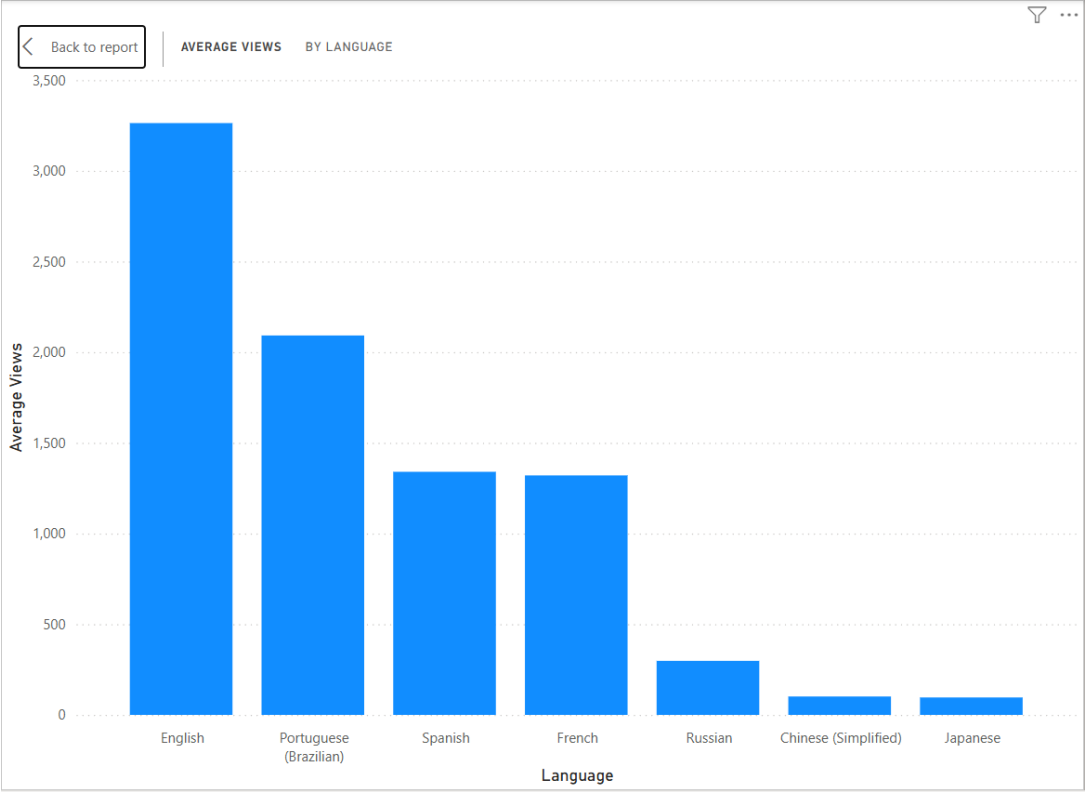
Category	Course Count
Arts and Humanities	120
Business	895
Computer Science	455
Data Science	448
Health	264
Information Technology	221
Language Learning	86
Math and Logic	22
Personal Development	68
Physical Science and Engineering	150
Social Sciences	83
Total	2812

Category	Course Count
Arts and Humanities	120
Philosophy	7
Music and Art	106
History	7
Business	895
Computer Science	455
Data Science	448
Health	264
Information Technology	221
Language Learning	86
Math and Logic	22
Personal Development	68
Physical Science and Engineering	150
Social Sciences	83
Total	2812

Question C3: Average Views by Category, Sub-category, and Language

- To analyze audience engagement across different dimensions, a DAX measure was created (Average Views = AVERAGE(FactCourses[Views_Clean])).
- This measure was then visualized across three distinct charts: a bar chart displaying average views by high-level Category, a bar chart displaying average views by specific Sub-category to identify niche popularity, and a column chart analyzing average viewership by Language to understand international audience distribution.
- All charts were sorted descending to highlight the highest-performing segments immediately.



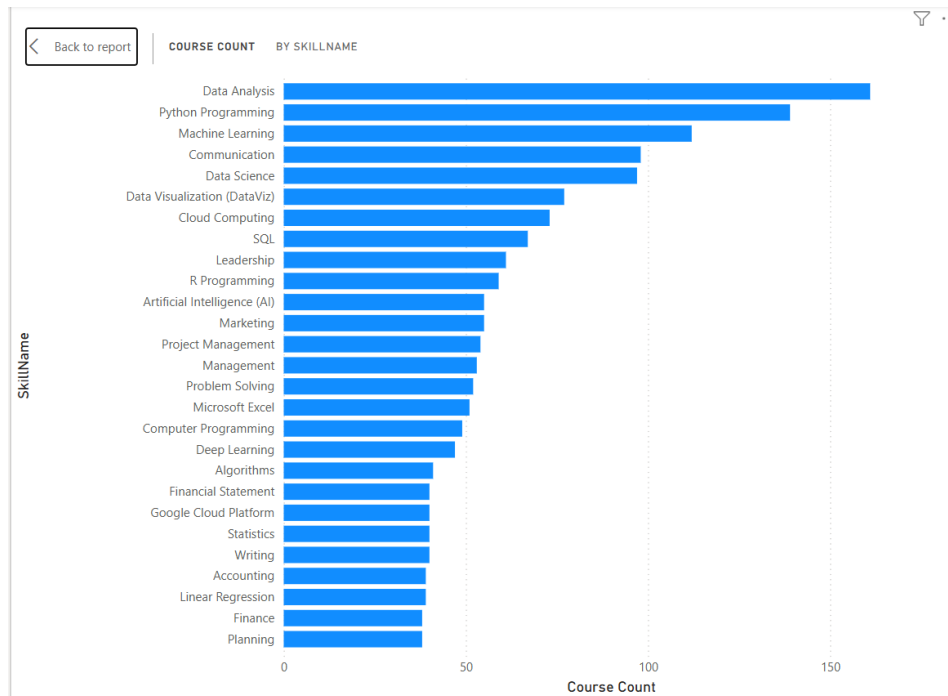


Question C4: Most Common Skills (Overall + Category-wise)

- Because the skills data was previously normalized into a dedicated DimSkills bridge table (Question B2), analyzing skill frequency was highly efficient.
- The existing Course Count measure was utilized alongside bidirectional cross-filtering to generate two visuals: a clustered bar chart displaying the Top 10 skills overall, achieved by applying a "Top N" visual-level filter based on the total course count, and a drill-down Matrix visual placing SkillName in a hierarchy beneath Category.
- Sorting the matrix values descending allows users to instantly identify the most frequently taught skills within any specific category grouping.

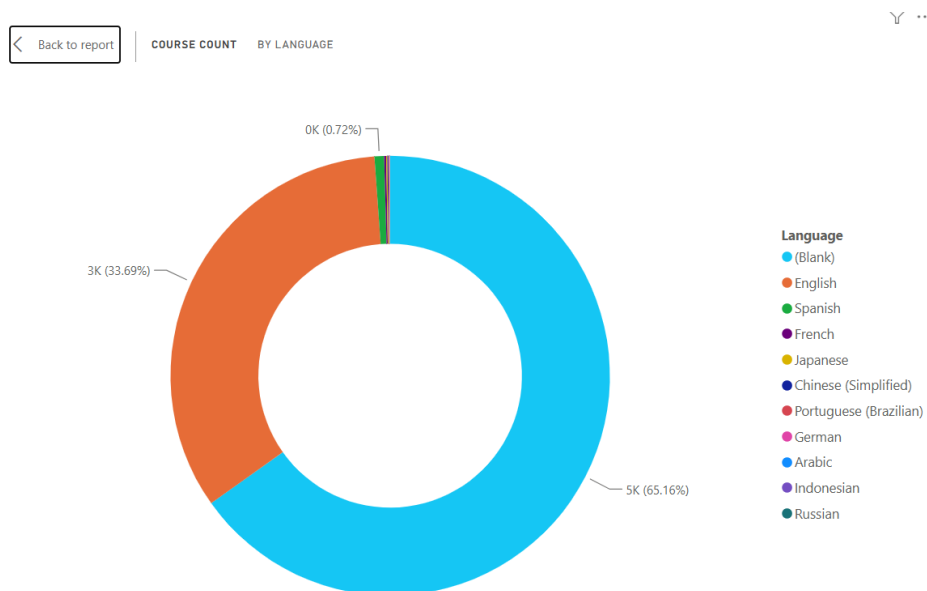
Category	Course Count
+ Business	895
+ Computer Science	455
+ Data Science	448
+ Health	264
+ Information Technology	221
+ Physical Science and Engineering	150
+ Arts and Humanities	120
+ Language Learning	86
+ Social Sciences	83
+ Personal Development	68
+ Math and Logic	22
Total	2812

Back to report	
Category	Course Count
+ Arts and Humanities	120
+ Business	895
	271
	624
1. Understand the meaning of organizational change management and what leaders need to understand regarding the reasons for change	1
2. Develop a successful change management team and define roles and responsibilities	1
3. Create a Change Management Communication Framework	1
3D Fashion Design	2
4. Develop communication processes during change management planning	1
5. Evaluate the results and outcomes of change through KPI analysis	1
80/20 Rule of Social Media	1
A/B Testing	3
â€¢ Agile Management	1
â€¢ Coaching	1
â€¢ Einflussnahme	1
â€¢ ProblemlÃ¶sung	1
â€¢ Scrum	1
Ã‰criture commerciale	1
Accounting	38
Accounting Analysis	1
Accounting Analytics	1
Accounting Concepts and Measurement	1
Accounting Cycle	3
Accounting for Liabilities and Equity	1
accounting software	1
Accounting Terminology	4
Accounts Payable and Payroll	1
Accounts Payable Management	2
Accounts receivable and cash receipts	1
Accounts Receivable Management	2
Total	2812



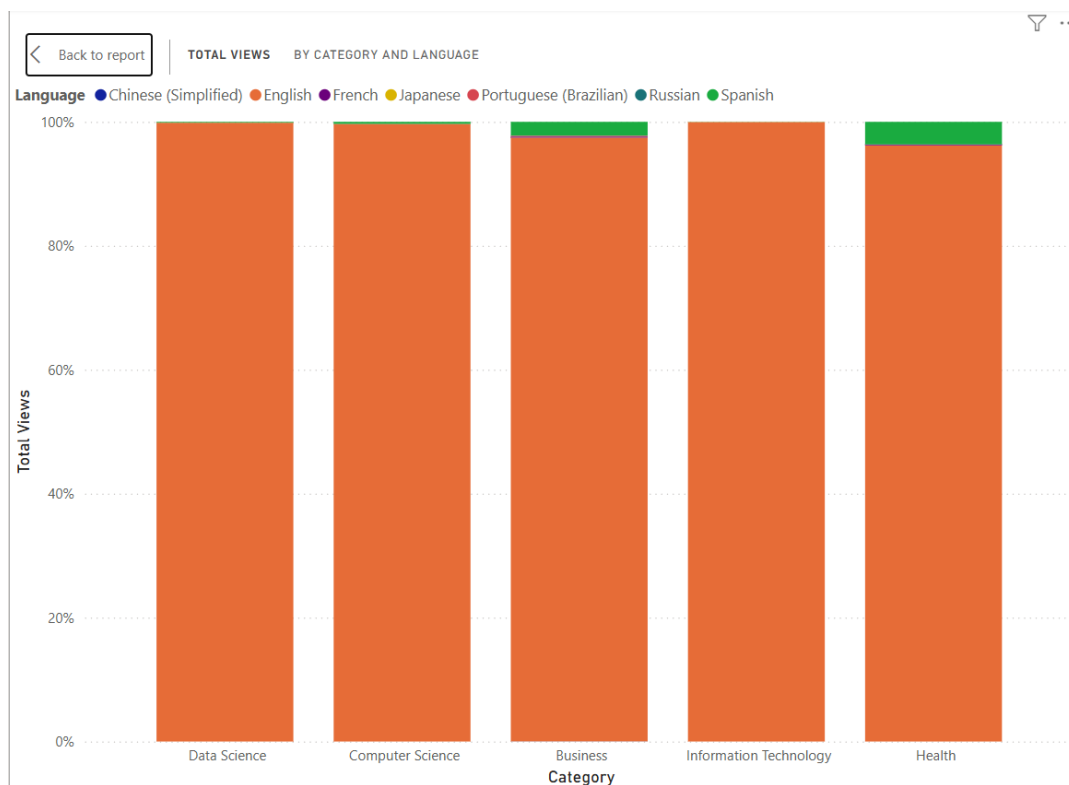
Question C5: Distribution of Languages Across Courses

- To evaluate the geographical and linguistic diversity of the course catalog, a Donut Chart was implemented to display the relative market share of each language.
- The visual utilizes the Course Count DAX measure aggregated by the clean attributes in the DimLanguage dimension table.
- Visual-level filtering was applied to omit blank records, providing an immediate understanding of which languages dominate the platform's offerings.



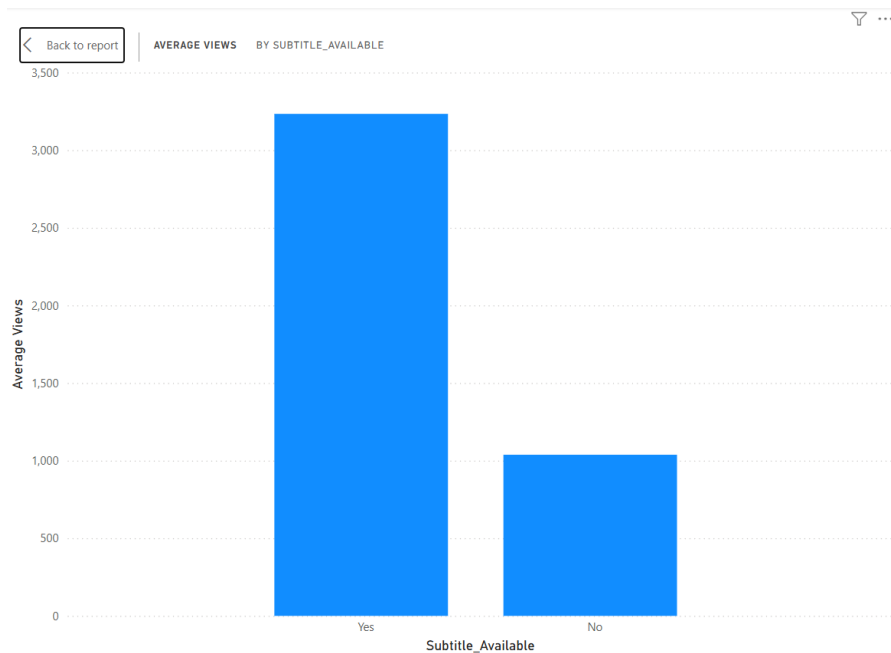
Question C6: Language Preferences for Top 5 Categories

- To analyze the dominant languages in the most popular subject areas, a new DAX measure was created (Total Views = SUM(FactCourses[Views_Clean])).
- This measure was visualized using a 100% Stacked Column Chart with Category on the X-axis and Language in the legend.
- A visual-level "Top N" filter was applied to restrict the view to only the Top 5 categories by total views.
- This effectively highlights the proportional language distribution for the platform's highest-traffic content.



Question C7: Subtitle Availability vs Views

- To investigate the impact of accessibility features on audience engagement, a Clustered Column Chart was created to compare the Average Views measure between courses with and without subtitles.
- To enable deeper drill-down analysis, a Category Slicer was added to the dashboard.
- This interactive element allows stakeholders to filter the subtitle comparison chart by specific disciplines, revealing whether the demand for subtitles fluctuates depending on the subject matter being taught.



Question C8: Top 3 Instructors Per Category & Sub-category (Static)

- To isolate the top-performing talent across the platform, a contextual ranking DAX measure (Instructor Rank) was developed using the RANKX and ALL functions.
- This measure evaluates the Avg Rating of an instructor against all other instructors within the exact same active filter context.
- A Matrix visual was constructed utilizing a row hierarchy of Category → Sub-category → Instructor.
- A visual-level filter was then applied to restrict the view to rows where Instructor Rank is less than or equal to 3, providing a static ledger of the top 3 instructors per discipline based on user evaluations.

< Back to report

Category

☐ Arts and Humanities

☐ Business

☐ Computer Science

☐ Data Science

☐ Health

☐ Information Technology

☐ Language Learning

☐ Math and Logic

☐ Personal Development

☐ Physical Science and Engineering

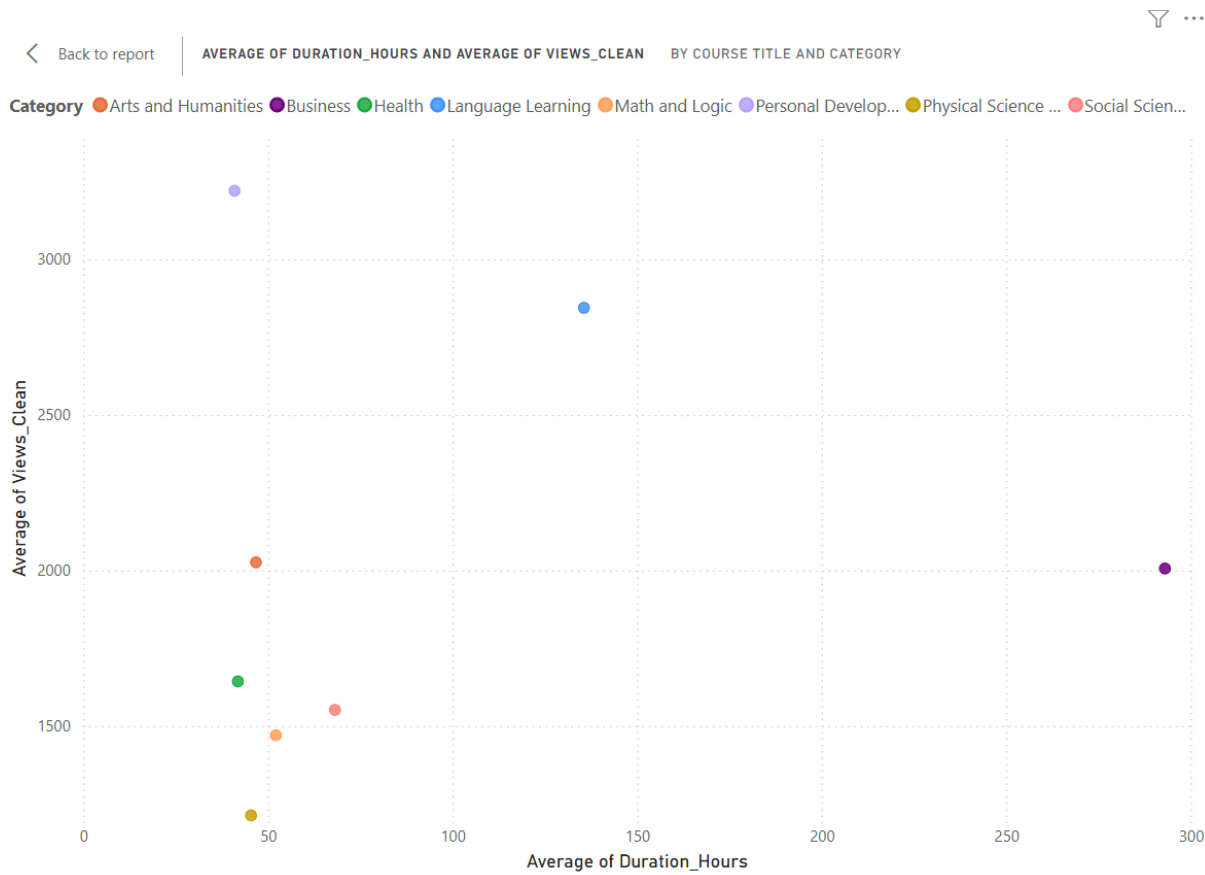
☐ Social Sciences

< Back to report

Category	Avg Rating	Instructor Rank
[-] Social Sciences	4.85	
[-] Education	4.92	
Barbara Oakley ,Beth Rogowsky ,Dr. Terrence Sejnowski ,David Joyner ,	5.00	1
Barbara Oakley	4.93	2
Brenda J. Allen	4.90	3
Dr. Shane Dixon	4.90	3
Emilia Gracia	4.90	3
Goodwill Academy	4.90	3
JoLynn Fiorentino	4.90	3
[+] Law	4.89	
[+] Governance and Society	4.81	
[+] Economics	4.78	
[+] Math and Logic	4.84	
[+] Data Science	4.84	
[+] Language Learning	4.83	
[+] Business	4.82	
[+] Health	4.80	
[+] Computer Science	4.80	
[+] Physical Science and Engineering	4.79	
[+] Arts and Humanities	4.78	
[+] Personal Development	4.78	
[+] Information Technology	4.77	
Total	4.81	

Question C9: Duration vs Views Relationship

- To analyze the correlation between course length and audience engagement, a Scatter Plot was generated.
- Duration_Hours was plotted on the X-axis against Views_Clean on the Y-axis, with individual data points representing each Course Title.
- The visual was color-coded by Category using the Legend field to identify cluster patterns among different disciplines.
- To ensure the visualization remained interpretable and was not skewed by extreme data points, visual-level filters were applied to remove statistical outliers (e.g., abnormally long courses or viral outliers with disproportionate viewership).



Question C10: Skill Variety vs Viewership

- To investigate whether a broader curriculum drives higher audience engagement, a DAX measure was formulated ($\text{Skill Variety} = \text{DISTINCTCOUNT}(\text{DimSkills}[\text{SkillName}])$) to quantify the number of unique skills taught within a given subject area.
- A Scatter Plot was deployed to correlate this new measure (X-axis) against viewership metrics (Y-axis) at the Sub-category granularity.
- The visual is color-coded by Category to identify if certain disciplines benefit more from skill diversification than others.
- The resulting distribution allows stakeholders to determine if teaching a wider variety of skills yields a proportional increase in user views.

Category Arts and ... Business Comput... Data Sci... Health Informati... Langua... Math an... Personal ... Physical ... Social S...

